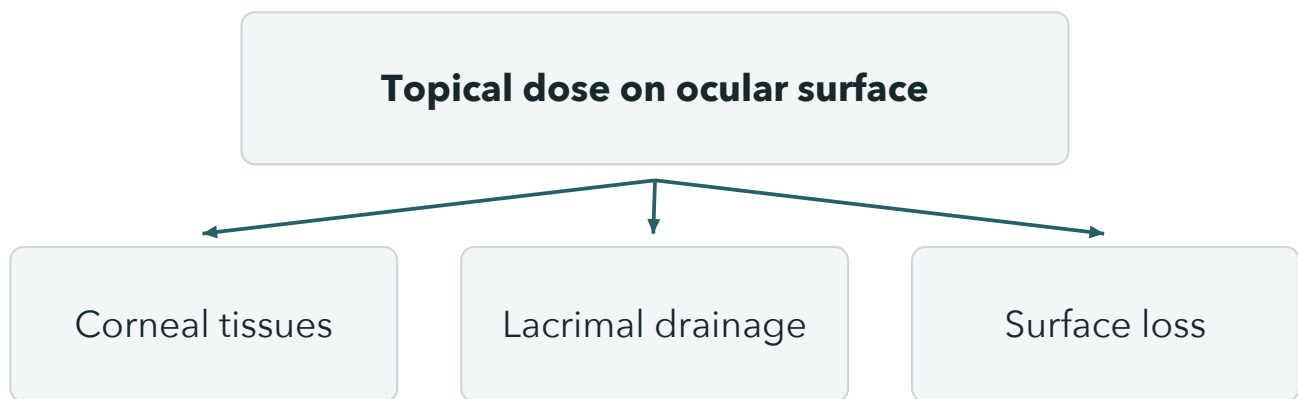


Ocular drug exposure and pharmacokinetics

Part 1 | An ocular dose has several exits

Local instillation does not guarantee exclusively local exposure.



Drainage can expose nasal mucosa and systemic circulation

A topical dose can enter corneal tissues, drain through the lacrimal route or be lost from the surface; proportions vary.

Route rather than chemical transformation

Nasolacrimal drainage exposes nasal mucosa and can contribute to systemic absorption. Punctal occlusion or eyelid closure can reduce that route for suitable drops. The mechanism is altered access, not renal acceleration or instant chemical inactivation of the drug.

Exposure is not the same as nominal dose

Delivery losses and tissue barriers separate the bottle concentration from the target concentration.

Step

Feature	Variable	Consequence
Surface residence	Tears, drainage, blinking	Contact time changes
Tissue access	Permeability and route	Target exposure changes
Elimination	Local and systemic clearance	Duration changes

A controlled comparison

If two formulations have equal drug concentration but different residence times, an exposure difference can arise before receptor binding. To attribute a changed effect to potency, first assess whether comparable amounts reached the target tissue. Nominal dose alone does not settle that comparison.

Loading and maintenance depend on different parameters

Distribution sets the amount needed to reach a target; clearance sets replacement rate.

One-compartment model

Quantity	Expression
Loading dose	Target concentration x V_d / F
Maintenance rate	Target concentration x clearance / F
F	Bioavailability fraction

Original example

With target concentration 2 mg/L, V_d 20 L and $F=1$, loading is 40 mg. If clearance is 3 L/hour, maintenance is 6 mg/hour. Doubling V_d doubles loading; halving clearance halves maintenance, with the other assumptions unchanged. These are model calculations, not a prescribed clinical regimen.

Steady state is a balance

A flat average concentration can conceal repeated peak-trough variation.

Repeated dosing model

- Drug is added at intervals
↓
- Drug is removed between doses
↓
- Accumulation continues until input and output balance
↓
- A repeating concentration pattern develops

Interpret the model

At linear pharmacokinetic steady state, average input matches average elimination. A loading dose can shorten the approach to the intended concentration range; it does not change intrinsic clearance. Nonlinear kinetics or changing physiology require a different analysis from the fixed-parameter model.

A concentration-effect change has several explanations

Distinguish delivery, kinetics and receptor response before naming resistance.

Observation

Feature	Candidate level	Next comparison
Lower tissue concentration	Access or clearance	Measure exposure
Same exposure, weaker response	Pharmacodynamics	Assess target response
Higher systemic adverse effect	Systemic access	Review drainage and route

Postgraduate reasoning

A route modification can change local and systemic exposure differently. Describe the specific compartment being measured and the sampling time. Without that information, the phrase drug concentration may conceal several biologically different quantities.